

A Counterexample to Removing the Invariant-Measure Hypothesis in the Fejer-to-AP Theorem

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Source and Question

The source paper is Anshu, Amrutam, and Karmakar, *Fejer property and Galois correspondence for groupoid C^* -algebras*, arXiv:2511.14474 [1]. Theorem 3.4 proves that if $\Gamma \ltimes X$ has the Fejer property and X admits a Γ -invariant measure, then Γ has the Haagerup–Kraus approximation property (AP). The following remark then appears.

Remark 3.5. We do not know if the assumption that μ is invariant can be removed in general.

isfies

$$\|M_{h_i}\|_{cb} = \max \left(\|M_{h_i^{(1)}}\|_{cb}, \|M_{h_i^{(2)}}\|_{cb} \right),$$

we see that $(h_i^{(j)})$ is a Fejér net for Γ_j .

It is natural to ask if $\Gamma \rtimes X$ having *Fejér* has any relation with the approximation property of the group Γ . We show that this is the case if X , in addition, admits an invariant measure μ .

Theorem 3.4. *Let X be a Γ -space. Assume that $\Gamma \rtimes X$ has Fejér property in our sense, and suppose that μ is Γ -invariant measure on X . Then Γ has property AP.*

Proof. Assume that $\Gamma \rtimes X$ has Fejér realized by the net $\{h_j\}$. Let μ be a Γ -invariant measure on X . For each $h_j \in C_c(\Gamma \rtimes X)$, define $\tilde{h}_j \in C_c(\Gamma)$ by

$$\tilde{h}_j(s) = \int_X h_j(s, x) d\mu(x), \quad s \in \Gamma.$$

Since μ is a Γ -invariant measure, the map $E_\mu : C(X) \rtimes_r \Gamma \rightarrow C_r^*(\Gamma)$ defined by $f\lambda(s) \rightarrow \mu(f)\lambda(s)$ is a Γ -equivariant conditional expectation (see [4, Exercise 4.1.4]). Letting $M_{\tilde{h}_j} = E_\mu \circ M_{h_j}$, we see that

$$\begin{aligned} \|M_{\tilde{h}_j}(a) - a\|_r &= \|M_{\tilde{h}_j}(a) - E_\mu(a)\|_r \\ &= \|E_\mu \circ M_{h_j}(a) - E_\mu(a)\|_r \leq \|M_{h_j}(a) - a\|_r \rightarrow 0. \end{aligned}$$

Also, note that $\|M_{\tilde{h}_j}\|_{cb} \leq \|M_{h_j}\|_{cb}$ for each j . Hence, $\sup_j \|M_{\tilde{h}_j}\|_{cb} < \infty$. Thus, Γ has AP. $\square$

Remark 3.5. We do not know if the assumption that μ is invariant can be removed in general.

We now show that when $\mathcal{G}$ has the *Fejér property*, the support of an element in $C_r^*(\mathcal{G})$ is contained in the subalgebra generated by an open set. In particular, the support of an element determines the subalgebra it belongs to. The following proposition makes it precise. The idea of the proof is essentially contained in [3, Theorem]. We sketch it nonetheless with all its details. Recall that a subset B of an étale groupoid is called a bisection if there is an open set U containing B such that $r : U \rightarrow r(U)$ and $s : U \rightarrow s(U)$ are homeomorphisms.

Proposition 3.6. *Assume that $\mathcal{G}$ has the Fejér property. Let U be an open subset of $\mathcal{G}$ and let $f \in C_r^*(\mathcal{G})$ be such that $\text{supp}(f) \subseteq U$. Then, $f \in \overline{C_c(U)}^r$.*

Proof. Since $\mathcal{G}$ has the Fejér property, there exists a net $(h_\alpha) \in C_c(\mathcal{G})$ satisfying

Figure 1: Theorem 3.4 and Remark 3.5 in arXiv:2511.14474, p. 7.

The answer is negative.

Proof Intuition

The invariant measure in Theorem 3.4 is used to average a groupoid Fejér net over the unit space and produce a Fejér net on the group itself. Without such an averaging functional, topologically amenable actions give immediate counterexamples: an exact group can act amenably on a compact space even when the group does not have AP.

The group $\text{SL}_3(\mathbb{Z})$ is the standard exact non-AP example. Its action on the flag manifold $\text{SL}_3(\mathbb{R})/P$, where P is a minimal parabolic subgroup, is topologically amenable because P is amenable. Adding a profinite left-translation factor makes the action free while preserving amenability. The resulting transformation groupoid has bounded Fejér property, but the acting group still lacks AP.

The Counterexample

Theorem 1. *There is a compact metrizable $\mathrm{SL}_3(\mathbb{Z})$ -space X such that the transformation groupoid $\mathrm{SL}_3(\mathbb{Z}) \ltimes X$ is second countable, Hausdorff, étale, principal, and has the bounded Fejer property, while $\mathrm{SL}_3(\mathbb{Z})$ does not have AP.*

Consequently, the invariant-measure hypothesis in Theorem 3.4 of [1] cannot be removed.

Proof. Let $\Gamma = \mathrm{SL}_3(\mathbb{Z})$ and let $G = \mathrm{SL}_3(\mathbb{R})$. Let $P < G$ be a minimal parabolic subgroup; concretely, one may take the upper triangular subgroup of G . The quotient

$$Y = G/P$$

is the compact metrizable full flag manifold.

The subgroup P is amenable. Hence the action $G \curvearrowright G/P$ is topologically amenable, and restricting the action to the discrete subgroup $\Gamma < G$ preserves topological amenability.

Let $Z = \widehat{\Gamma}$ be the profinite completion of Γ , with Γ acting by left translation. Since Γ is finitely generated and residually finite, Z is compact metrizable and the natural map $\Gamma \rightarrow Z$ is injective. The left-translation action on Z is free: if $gz = z$ for some $z \in Z$, then $g = e$ in the group Z , hence $g = e$ in Γ .

Set

$$X = Y \times Z$$

with the diagonal Γ -action. The action remains topologically amenable: if $m_i : Y \rightarrow \mathrm{Prob}(\Gamma)$ are continuous approximately equivariant maps witnessing amenability of $\Gamma \curvearrowright Y$, then

$$(y, z) \mapsto m_i(y)$$

witnesses amenability of $\Gamma \curvearrowright X$. It is also free because the second coordinate action is free. Therefore $\Gamma \ltimes X$ is a second countable Hausdorff étale principal transformation groupoid.

It remains only to record that this groupoid has the bounded Fejer property. For a topologically amenable action of a discrete group on a compact space, there is a net $h_i \in C_c(\Gamma \ltimes X)$ of positive-type functions such that $h_i \rightarrow 1$ uniformly on compact subsets of $\Gamma \ltimes X$. The pointwise multipliers M_{h_i} are completely positive contractions on

$$C_r^*(\Gamma \ltimes X) \cong C(X) \rtimes_r \Gamma.$$

For an element $a = \sum_{s \in F} a_s u_s \in C_c(\Gamma \ltimes X)$, where $F \subset \Gamma$ is finite, uniform convergence on $F \times X$ gives

$$\begin{aligned} \|M_{h_i}(a) - a\| &\leq \sum_{s \in F} \|(h_i(s, \cdot) - 1)a_s\|_{C(X)} \\ &\leq \sum_{s \in F} \|a_s\|_{C(X)} \sup_{(s, x) \in F \times X} |h_i(s, x) - 1| \longrightarrow 0. \end{aligned}$$

Since $C_c(\Gamma \ltimes X)$ is dense in $C_r^*(\Gamma \ltimes X)$ and the multipliers have cb-norm at most 1, the convergence holds for every element of the reduced crossed product. Thus $\Gamma \ltimes X$ has the bounded Fejer property.

On the other hand, $\mathrm{SL}_3(\mathbb{Z})$ does not have AP, by the Lafforgue–de la Salle theorem as cited in [1] and in the standard literature [3]. Therefore the implication

$$\Gamma \ltimes X \text{ has Fejer property} \implies \Gamma \text{ has AP}$$

is false without an invariant-measure hypothesis. This is precisely the proposed removal in Remark 3.5. $\square$

Remark 1. *The example necessarily has no Γ -invariant probability measure on X . Indeed, if such a measure existed, Theorem 3.4 of the source paper would apply to the Fejer groupoid $\Gamma \ltimes X$ and would force $\mathrm{SL}_3(\mathbb{Z})$ to have AP, contradicting the final paragraph of the proof.*

Verification Notes

No numerical computation is used. The proof depends on the following standard inputs:

1. $\mathrm{SL}_3(\mathbb{Z})$ is finitely generated, residually finite, and does not have AP.
2. If P is a minimal parabolic subgroup of $\mathrm{SL}_3(\mathbb{R})$, then P is amenable and the action on G/P is topologically amenable.
3. A topologically amenable action of a discrete group on a compact space gives compactly supported positive-type Herz–Schur multipliers converging to 1, hence the bounded Fejer property for the transformation groupoid.

Scope and Novelty Check

This packet answers only Remark 3.5/Theorem 3.4 of the source paper. It does not address Question 1 of Fuller–Karmakar about Fourier recovery for arbitrary groupoids, and it does not challenge the source paper’s Galois correspondence or bimodule theorems.

The run’s cheap indexes were searched for arXiv:2511.14474, the source title, “Fejer”, “invariant measure”, “removed in general”, “amenable action”, “transformation groupoid”, “SL3”, and “AP”. No prior packet for this paper or route was found. A bounded web search on 2026-07-18 for exact and nearby phrases including “Fejer property invariant measure removed groupoid”, “Gamma rtimes X Fejer property invariant measure AP”, and “SL3(Z) approximation property exact amenable action” found the source paper and standard background on exact non-AP groups and amenable actions, but no existing answer to Remark 3.5.

Human Review Recommendation

Recommended verdict: likely valid counterexample. Reviewers should mainly check the standard amenability-to-positive-type-multiplier passage for the specific Fejer definition in the source paper. The obstruction to dropping the invariant-measure hypothesis is otherwise direct: $\Gamma \ltimes X$ has bounded Fejer property while $\Gamma = \mathrm{SL}_3(\mathbb{Z})$ lacks AP.

References

- [1] Anshu, Tattwamasi Amrutam, and Pradyut Karmakar, *Fejer property and Galois correspondence for groupoid C^* -algebras*, arXiv:2511.14474, 2025.
- [2] Claire Anantharaman-Delaroche and Jean Renault, *Amenable groupoids*, Monographies de L’Enseignement Mathematique 36, 2000.
- [3] Uffe Haagerup and Tim de Laat, *Simple Lie groups without the approximation property*, Duke Mathematical Journal 162 (2013), no. 5, 925–964.

- [4] Nathaniel P. Brown and Narutaka Ozawa, *C^* -algebras and finite-dimensional approximations*, Graduate Studies in Mathematics 88, American Mathematical Society, 2008.